HDOC DAY-11 ACTIVITY REPORT

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Batch: 15

Menu-Driven C Program Using Switch-Case

Submitted to: Dr. Christalin Nelson

Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. Find the sum of squares of its digits using a while loop. Example: Number = 123, Sum = 14.

2. Find the sum of cubes of its digits using a for loop. Example: Number = 123, Sum = 36.

3. Find the difference between the sum of cubes and the sum of squares of its digits using a do-while loop. Example: Number = 123, Difference = 22.

Algorithm

1. Start the program.

2. Input a positive integer n.

3. Display a menu with three operations and read the user's choice.

4. Use switch-case to execute the selected operation.

5. For choice 1, repeatedly extract each digit using modulus 10, add the square of the digit to sum, and remove the digit using integer division. Use a while loop.

6. For choice 2, extract each digit and add its cube to the sum. Use a for loop.

7. For choice 3, use a do-while loop to calculate both the sum of squares and the sum of cubes, then compute: Difference = Sum of Cubes - Sum of Squares.

8. Display the calculated result.

9. Stop.

Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	n = 123, choice = 1	Sum of squares = 14	14	Pass
2	n = 123, choice = 2	Sum of cubes = 36	36	Pass
3	n = 123, choice = 3	Difference = 22	22	Pass

Program Screenshots

The C program is shown below using the provided screenshots. No plain-text code has been added.

```
  UW PICO 5.09                                File: day12.cp

#include <stdio.h>

int main()
{
    int n, choice, k;
    int first, last, sum, product;
    int digit, lesser, equal, greater;
    int temp;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    do
    {
        printf("\n===== MENU =====\n");
        printf("1. Sum and Product of First and Last Digits\n");
        printf("2. Count Digits Lesser, Equal and Greater than k\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                temp = n;

                last = temp % 10;

                while (temp >= 10)
                {
                    temp = temp / 10;
                }

                first = temp;

                sum = first + last;
                product = first * last;

                printf("First digit = %d\n", first);
                printf("Last digit = %d\n", last);
                printf("Sum = %d\n", sum);
                printf("Product = %d\n", product);

                break;

            case 2:
                printf("Enter digit k (0-9): ");
                scanf("%d", &k);

                lesser = 0;
                equal = 0;
                greater = 0;
                temp = n;

                while (temp > 0)
^G Get Help      ^O WriteOut      ^R Read File     ^Y Prev Pg       ^K Cut Text      ^C Cur Pos
^X Exit          ^J Justify       ^W Where is      ^V Next Pg       ^U UnCut Text    ^T To Spell
```

```
  UW PICO 5.09                                File: day12.cp

                while (temp >= 10)
                {
                    temp = temp / 10;
                }

                first = temp;

                sum = first + last;
                product = first * last;

                printf("First digit = %d\n", first);
                printf("Last digit = %d\n", last);
                printf("Sum = %d\n", sum);
                printf("Product = %d\n", product);

                break;

            case 2:
                printf("Enter digit k (0-9): ");
                scanf("%d", &k);

                lesser = 0;
                equal = 0;
                greater = 0;
                temp = n;

                while (temp > 0)
                {
                    digit = temp % 10;

                    if (digit < k)
                        lesser++;
                    else if (digit == k)
                        equal++;
                    else
                        greater++;

                    temp = temp / 10;
                }

                printf("Count_lesser = %d\n", lesser);
                printf("Count_equal = %d\n", equal);
                printf("Count_greater = %d\n", greater);

                break;

            case 3:
                printf("Program terminated.\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 3);
^G Get Help      ^O WriteOut      ^R Read File     ^Y Prev Pg       ^K Cut Text      ^C Cur Pos
^X Exit          ^J Justify       ^W Where is      ^V Next Pg       ^U UnCut Text    ^T To Spell
```

```
UW PICO 5.09                                   File: day12.cp

                digit = temp % 10;

                if (digit < k)
                    lesser++;
                else if (digit == k)
                    equal++;
                else
                    greater++;

                temp = temp / 10;
            }

            printf("Count_lesser = %d\n", lesser);
            printf("Count_equal = %d\n", equal);
            printf("Count_greater = %d\n", greater);

            break;

        case 3:
            printf("Program terminated.\n");
            break;

        default:
            printf("Invalid choice! Please try again.\n");
    }

    } while (choice != 3);

    return 0;
}

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```

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Execution / Output

```
[gurveer@Gurveers-MacBook-Air ~ % nano day12.cp
[gurveer@Gurveers-MacBook-Air ~ % clang day12.cp -o day12
[gurveer@Gurveers-MacBook-Air ~ % ./day12
Enter a positive integer: 12345

===== MENU =====
1. Sum and Product of First and Last Digits
2. Count Digits Lesser, Equal and Greater than k
3. Exit
Enter your choice: 1
First digit = 1
Last digit = 5
Sum = 6
Product = 5

===== MENU =====
1. Sum and Product of First and Last Digits
2. Count Digits Lesser, Equal and Greater than k
3. Exit
Enter your choice: 2
Enter digit k (0-9): 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2

===== MENU =====
1. Sum and Product of First and Last Digits
2. Count Digits Lesser, Equal and Greater than k
3. Exit
Enter your choice: 1
First digit = 1
Last digit = 5
Sum = 6
Product = 5

===== MENU =====
1. Sum and Product of First and Last Digits
2. Count Digits Lesser, Equal and Greater than k
3. Exit
Enter your choice: 
```

Result

The menu-driven C program was compiled and executed successfully. The required operations were tested and the obtained results matched the expected results.